ASSIGNMENT 1

JavaScript Programs and Solutions

Q1. Personalized Login Greeting

```
// Q1_PersonalizedLoginGreeting.js
const userName = "Anurag";
const currentHour = new Date().getHours();

if (currentHour < 12) {
    console.log(`Good Morning ${userName}!`);
} else if (currentHour <= 17) {
    console.log(`Good Afternoon ${userName}!`);
} else {
    console.log(`Good Evening ${userName}!`);
}
```

Q2. Multi-Type Data Summary

```
// Q2_MultiTypeDataSummary.js
const userName = "Anurag";
const age = 21;
const isStudent = true;
const subjects = ["JavaScript", "Java"];
const student = { city: "Etawah" };
const middleName = null;
let nickname;

const report = [
    { Label: "Name", Value: userName, Type: typeof userName },
    { Label: "Age", Value: age, Type: typeof age },
    { Label: "Student", Value: isStudent, Type: typeof isStudent },
    { Label: "Subjects", Value: subjects.join(", "),
      Type: Array.isArray(subjects) ? "array" : typeof subjects },
    { Label: "Student object", Value: JSON.stringify(student),
      Type: typeof student },
    { Label: "Middle name", Value: String(middleName),
      Type: middleName === null ? "null" : typeof middleName },
    { Label: "Nickname", Value: String(nickname), Type: typeof nickname }
];

console.table(report);
```

Q3. Monthly Expense Tracker

```
// Q3_MonthlyExpenseTracker.js
const expenses = [4500, 1200, 7000, 1500, 800];
// Order: food, travel, rent, bills and leisure

let total = 0;

for (const expense of expenses) {
    total += expense;
}

const average = total / expenses.length;
const tax = total * 0.10;
total += tax;

console.log("Total before tax: Rs. " + (total - tax).toFixed(2));
console.log("Average expense: Rs. " + average.toFixed(2));
console.log("Tax (10%): Rs. " + tax.toFixed(2));
console.log("Final amount: Rs. " + total.toFixed(2));
```

Q4. Academic Performance Evaluator

```
// Q4_AcademicPerformanceEvaluator.js
const marks = [88, 76, 91, 67, 82]; // Marks out of 100

let totalMarks = 0;

for (const mark of marks) {
    totalMarks += mark;
}

const average = totalMarks / marks.length;
const percentage = (totalMarks / (marks.length * 100)) * 100;
const hasFailed = marks.some(mark => mark < 35);

console.log("Average marks: " + average.toFixed(2));
console.log("Percentage: " + percentage.toFixed(2) + "%");

if (hasFailed || percentage < 50) {
    console.log("Detained");
} else if (percentage >= 85) {
    console.log("Promoted with Distinction");
} else {
    console.log("Promoted");
}
```

Q5. Weather Activity Planner

```
// Q5_WeatherActivityPlanner.js
const temperature = 12;
const isRaining = false;
const windSpeed = 25;

if (isRaining) {
    console.log("Stay indoors with hot coffee.");
} else if (temperature > 35) {
    console.log("Go swimming.");
} else if (temperature < 15 && windSpeed > 20) {
    console.log("Too cold and windy — stay home.");
} else if (!isRaining && (temperature >= 15 || windSpeed <= 20)) {
    console.log("Perfect day for a walk.");
} else {
    console.log("Check the weather conditions.");
}
```

Q6. Progressive Discount System

```
// Q6_ProgressiveDiscountSystem.js
const purchaseAmount = 6500;
let discountPercent = 0;

if (purchaseAmount >= 10000) {
    discountPercent = 25;
} else if (purchaseAmount >= 5000) {
    discountPercent = 15;
} else if (purchaseAmount >= 2000) {
    discountPercent = 5;
}

const discountAmount = Math.round(
    purchaseAmount * discountPercent / 100
);
const finalPrice = Math.round(purchaseAmount - discountAmount);

console.log("Original total: Rs. " + purchaseAmount);
console.log("Discount: " + discountPercent + "%");
console.log("Discount amount: Rs. " + discountAmount);
console.log("Final price: Rs. " + finalPrice);
```

Q7. Smart Guessing Game

```
// Q7_SmartGuessingGame.js
const secretNumber = Math.floor(Math.random() * 50) + 1;
const userGuess = 24; // Change this value to test the game

if (userGuess < 1 || userGuess > 50) {
    console.log("Please enter a number from 1 to 50.");
} else if (userGuess === secretNumber) {
    console.log("Correct guess!");
} else if (Math.abs(userGuess - secretNumber) <= 3) {
    console.log("Very close!");
} else if (userGuess > secretNumber) {
    console.log("Too high");
} else {
    console.log("Too low");
}

// This line helps us test the program.
console.log("Secret number: " + secretNumber);
```

Q8. Employee Salary Projection

```
// Q8_EmployeeSalaryProjection.js
const currentSalary = 30000;
const incrementRate = 8; // 8% yearly increment
let salary = currentSalary;
const salaryDetails = [];

for (let year = 1; year <= 5; year++) {
    salary += salary * incrementRate / 100;
    salary = Math.round(salary * 100) / 100;

    salaryDetails.push({
        Year: year,
        "Monthly Salary (Rs.)": salary.toFixed(2)
    });
}

console.table(salaryDetails);
```

Q9. Random Math Quiz Generator

```
// Q9_RandomMathQuizGenerator.js
const num1 = Math.floor(Math.random() * 20) + 1;
const num2 = Math.floor(Math.random() * 20) + 1;
const operators = ["+", "-", "*", "/"];
const operator = operators[
    Math.floor(Math.random() * operators.length)
];

let answer;

switch (operator) {
    case "+":
        answer = num1 + num2;
        break;
    case "-":
        answer = num1 - num2;
        break;
    case "*":
        answer = num1 * num2;
        break;
    case "/":
        answer = num1 / num2;
        break;
}

console.log("Question: " + num1 + " " + operator + " " + num2);
console.log("Correct answer: " +
    (operator === "/" ? answer.toFixed(2) : answer));
```

Q10. Citizen Eligibility Validator

```
// Q10_CitizenEligibilityValidator.js
const age = 19;
const isCitizen = true;

if (isCitizen && age >= 21) {
    console.log("Eligible for all services.");
} else if (isCitizen && age >= 18 && age <= 20) {
    console.log("Eligible to vote only.");
} else if (!isCitizen && age >= 18) {
    console.log("Only age criteria met.");
} else {
    console.log("Not eligible yet.");
}
```